



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
PRE-SCORE
11-11-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

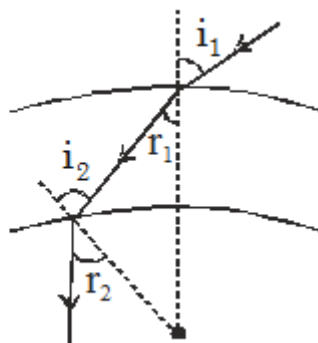
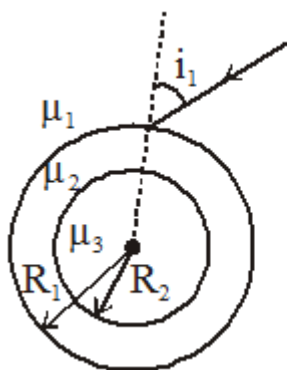
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options A, B, C and D. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Consider two spherical boundaries of radii R_1 and R_2 separating mediums of refractive indices μ_1 , μ_2 and μ_3 as shown. A light ray is incident at angle i_1 as shown. Light after 1st refraction strikes the boundary of radius R_2 separating mediums μ_2 & μ_3 where its angle of incidence is assumed to be i_2 as shown in diagram. Identify the correct relation between i_1 and i_2 .



(A) $\mu_1 \sin i_1 = \mu_2 \sin i_2$

(B) $\mu_1 R_1 \sin i_1 = \mu_2 R_2 \sin i_2$

(C) $\mu_1 R_2 \sin i_1 = \mu_2 R_1 \sin i_2$

(D) None of these

Space for Rough Work

2. A point object is placed at a distance of 62 cm from a concave mirror of focal length 20 cm. A slab of the thickness 6 cm and refractive index $\mu = 3/2$ is placed in between object and mirror at a distance 1.5 cm from mirror. The distance of final image from mirror is :
 (A) 29.5 cm (B) 32 cm (C) 33.5 cm (D) 30.5 cm
3. A thin mica sheet ($\mu = 1.6$) is used to cover one slit of Young's double slit arrangement. The central fringe is now formed at the previous seventh bright fringe. If $\lambda = 5500\text{\AA}$, the thickness of the mica sheet is :
 (A) $1.2\text{ }\mu\text{m}$ (B) $6.41\text{ }\mu\text{m}$ (C) 6.4 mm (D) $2.4\text{ }\mu\text{m}$
4. Two particles of mass m and $2m$ moving in opposite direction collide head on. They have same de-Broglie wavelength before collision. After the collision :-
 (A) If $e = 1$, de-Broglie wavelength of m is greater than that of $2m$.
 (B) If $e = 1$, de-Broglie wavelength of $2m$ is greater than that of m .
 (C) The de-Broglie wavelength of m increases if $e = 1$.
 (D) The de-Broglie wavelength of $2m$ remains same if $e = 1$.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. According to Bohr's theory, for a hydrogen-like atom of atomic number Z , the :-
- (A) Radius of an orbit is inversely proportional to Z .
- (B) Radius of an orbit is directly proportional to Z^2 .
- (C) Energy of electron in an orbit is directly proportional to Z^2 .
- (D) Energy of the electron in an orbit is directly proportional to Z .
6. In a hypothetical hydrogen-like atom, the wavelength in $\text{\AA}$ for the spectral lines for transitions from $n = p$ to $n = 1$ are given by $\lambda = \frac{1500p^2}{p^2 - 1}$, where $p = 2, 3, 4, \dots$ (Take : $hc = 12400 \text{ eV}\cdot\text{\AA}$) Choose the correct option(s) :
- (A) The ratio of wavelength of the least energetic and most energetic photon of this series is $\frac{4}{3}$
- (B) Energy of electron in ground state is 8.27 eV.
- (C) Wavelength of 1st Balmer line is approximately 12800 $\text{\AA}$.
- (D) Photon of wavelength 3000 $\text{\AA}$ will be absorbed by atom in ground state.

Space for Rough Work

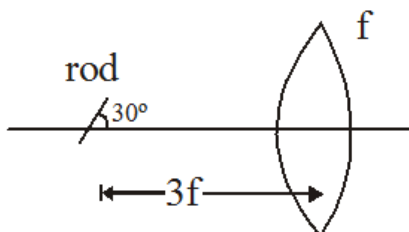
7. Radon decays to polonium according to the decay scheme given by ${}^{222}_{88}\text{Rn} \xrightarrow[3.8 \text{ days}]{\alpha\text{-decay}} {}^{218}_{84}\text{Po}$. State which of the following statement(s) will be true in respect of this decay : (Given : $\ln 2 = 0.693$)
- (A) If the original mass of radon in a closed container was 1mg, the amount of radon left after 11.4 days will be 0.125 mg.
- (B) Activity of radon after 7.6 days will be $\frac{A_0}{4}$ where A_0 is original activity of sample at $t = 0$.
- (C) If the original number of radon in a closed container was 8 atoms, the number of radon left after 7.6 day will be 2 atoms.
- (D) Po is more stable than Rn .
8. A partition divides a container having insulated walls into two compartments I and II. The same gas fill the two compartments whose initial parameters are given. The partition is conducting wall which can move freely without friction. Which of the following statement(s) is/are correct, with reference to the final equilibrium position ?

P, V, T I	$2P, 2V, T$ II
----------------	-------------------

- (A) The pressure in the two compartments are equal.
- (B) Volume of compartment I is $\frac{3V}{5}$.
- (C) Volume of compartment II is $\frac{3V}{5}$.
- (D) Final pressure in compartment I is $\frac{5P}{3}$.

Space for Rough Work

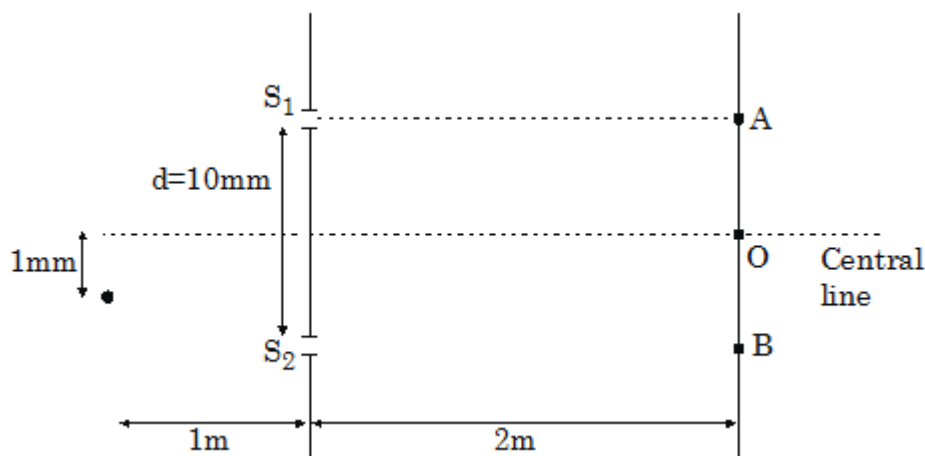
9. If a solid sphere and a hollow sphere of the same material and radius are heated to the same temperature. Choose the correct statement(s) :
- (A) The initial rate of heat radiated per unit time is same.
- (B) The initial rate of heat absorbed per unit time is same.
- (C) The rate of cooling will be same for both the spheres.
- (D) The two spheres will have same temperature at any instant.
10. A thin convex lens of focal length f is having small rod at it's optical axis at $3f$ distance in front of it. [angular velocity given is about axis $\perp$ to plane of paper]



- (A) Image of rod is inclined at $\tan^{-1} \left(\frac{2}{\sqrt{3}} \right)$ angle with optical axis.
- (B) If rod is rotated with angular velocity $\omega = 7$ rad/sec it's image will start to rotate with angular velocity 8 rad/s
- (C) If lens is rotated with angular velocity 2 rad/sec then image of rod will rotate with angular velocity 15 rad/s
- (D) length of image of rod is smaller than length of the rod.

Space for Rough Work

11. In given young's double slit experiment a point sources of wavelength $\lambda = 500 \text{ nm}$ is placed slightly off the central axis. Point A and B are in front of upper and lower slit respectively. If intensity from upper & lower slits is I_0 each on the screen, then



- (A) Intensity at A, O and B is $4I_0$, $2I_0$ and $4I_0$ respectively.
- (B) 70th maxima is formed at B
- (C) 30th maxima is formed at A
- (D) If a glass slab of thickness 0.02 mm and refractive index $\mu = 3/2$ is kept in front of upper slit then central maxima formed exactly at O.
12. A solid object has density ρ , mass M and coefficient of linear expansion α . Assume that $\frac{dU}{dT}$ at constant pressure is equal to $\frac{dU}{dT}$ at constant volume. Here U is internal energy and T is temperature. C_p and C_v are heat capacities at constant pressure and volume respectively.
- (A) $C_p - C_v = P \frac{dV}{dT}$
- (B) $\frac{dV}{dT} = 3\alpha \frac{M}{\rho}$
- (C) $C_p - C_v = 3\alpha P \frac{M}{\rho}$
- (D) $C_p - C_v = \alpha P \frac{M}{\rho}$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

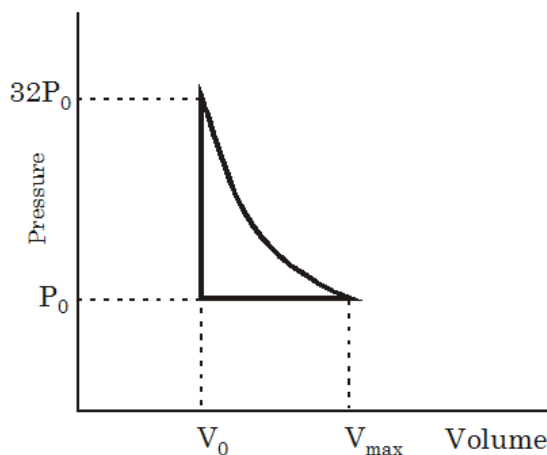
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A uniform rod is rotating about one of its end with a constant angular velocity ω in gravity free space. The ratio of strain produced in two halves is k ($k > 1$). The value of k is.
2. A long cylindrical vessel having a linear coefficient of expansion α is filled with a liquid upto a certain level. On heating, it is found that the height of liquid in the cylinder remains the same. The coefficient of volume expansion of liquid is given by $\beta \times 10^{-\gamma}$. Fill $\gamma + \beta$. (β & γ are smallest possible integers)
[Given : $\alpha = 5 \times 10^{-7}/^{\circ}\text{C}$]
3. Light of wavelength λ from a small 0.5 mW He-Ne laser source is used in the school laboratory to shine on perfectly absorbing surface of a space craft of mass 1000 kg. The time needed for the space craft to reach a velocity of 1 km/s from rest is given by $\alpha \times 10^{\beta}$ sec in scientific notation. Find the value of $(\beta - 2\alpha)$.
4. Consider a rod of length 1m. One of the ends of the rod is maintained at 100°C . The other end is at 0°C . At the distance of 25 cm from the end at higher temperature, there is a heat source giving heat at a constant rate of 80 W to the rod. The area of the rod is 100 cm^2 and coefficient of thermal conductivity is 50 W/mK . What is the temperature of the heat source ? (in $^{\circ}\text{C}$)

Space for Rough Work

5. A simple heat engine consists of a piston in a cylinder filled with an ideal monatomic gas. Initially the gas in the cylinder is at a pressure P_0 and volume V_0 . The gas is slowly heated at constant volume. Once the pressure reaches $32P_0$ the piston is released, allowing the gas to expand so that no heat neither enters nor escapes the gas as the piston moves. Once the pressure has returned to P_0 the outside of the cylinder is cooled back to the original temperature, keeping the pressure constant. For the monatomic ideal gas you should assume that the specific heat capacity at constant volume is given by $C_V = \frac{3}{2}nR$, where n is the number of moles of the gas present and R is the ideal gas constant. Let $V_{\max}$ be the maximum volume achieved by the gas during the cycle. What is $\frac{V_{\max}}{V_0}$.



6. The principal section of glass prism is an isosceles ΔPQR with $PQ = PR$. The face PR is silvered. A ray is incident perpendicularly on face PQ and after two reflections it emerges from base QR , normal to it. If the angle (in degree) of the prism is x . Find $\frac{x}{4}$.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

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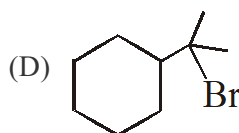
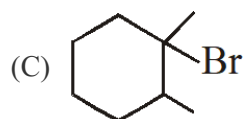
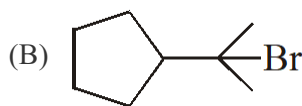
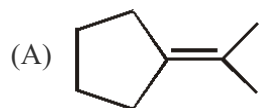
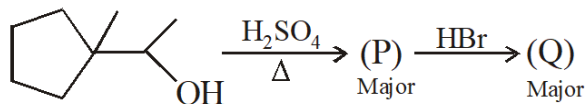
1. Calculate change in internal energy for 2 moles of an ideal gas when temperature rises from 27°C to 127°. [Given : $C_{p,m} \text{ (JK}^{-1}\text{mol}^{-1}) = 13.314 + 0.04 T$]
(A) 3800 J (B) 1900 J (C) 1028 J (D) 5462.8 J

2. The standard enthalpy of atomisation of $\text{PCl}_3(\text{g})$ is 195 Kcal/mol. What will be the standard enthalpy of atomisation of $\text{PCl}_5(\text{g})$, if the bond dissociation energies of axial P-Cl bonds in $\text{PCl}_5(\text{g})$ are 10% lesser and the bond dissociation energies of equatorial P-Cl bonds in $\text{PCl}_5(\text{g})$ are 10% higher than the bond dissociation energies of P-Cl bonds in $\text{PCl}_3(\text{g})$.
(A) 195 Kcal/mol (B) 325 Kcal/mol
(C) 331.5 Kcal/mol (D) -500 Kcal/mol

Space for Rough Work

3. FeCr_2O_4 (chromite) is converted to $\text{K}_2\text{Cr}_2\text{O}_7$ by following steps :
 Chromite $\xrightarrow{\text{I}}$ $\text{Na}_2\text{CrO}_4 \xrightarrow{\text{II}}$ $\text{Na}_2\text{Cr}_2\text{O}_7 \xrightarrow{\text{III}}$ $\text{K}_2\text{Cr}_2\text{O}_7$; I, II and III are :
 (A) I - $\text{Na}_2\text{CO}_3/\text{air}, \Delta$; II - H^+ ; III - KCl
 (B) I - $\text{NaOH}/\text{air}, \Delta$; II - OH^- , Δ ; III - HCl, Δ
 (C) I - $\text{NaOH}/\text{air}, \Delta$; II - C , Δ ; III - Fe, Δ
 (D) I - $\text{Conc. H}_2\text{SO}_4$, Δ ; II - NH_4Cl , Δ ; III - C , Δ

4. The final major product (Q) of the given sequence of reaction :



Space for Rough Work

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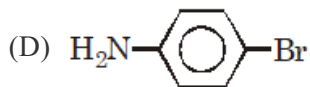
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5. Select correct statement for an ideal gas undergoing reversible or irreversible adiabatic process from same initial state :
- (A) For same final pressure, final temperature is more in irreversible process
- (B) For same final volume, final temperature is more in irreversible process
- (C) For same final temperature, final pressure is more in irreversible process
- (D) For same final temperature, final volume is more in irreversible process
6. Select the correct statement(s)
- (A) Heat is property of system
- (B) $\Delta H = \Delta U + \Delta (PV)$ can be used for both ideal & real gas
- (C) $\Delta S = nC_{p,m} \ln \frac{T_2}{T_1}$ can be used for both ideal & real gas in isobaric process
- (D) $\Delta S = nR \ln \frac{V_2}{V_1}$ can be used for both ideal & real gas in isothermal process
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Space for Rough Work

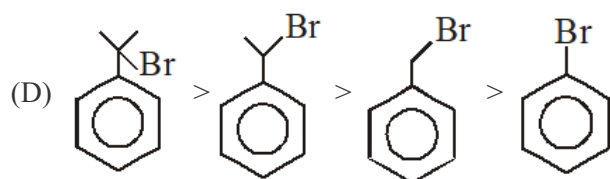
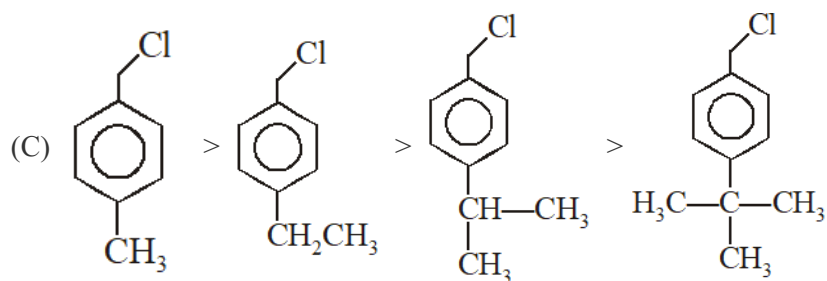
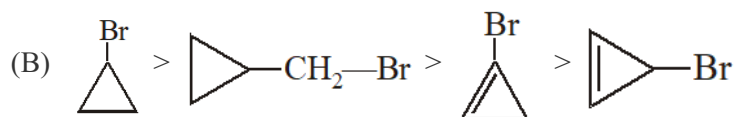
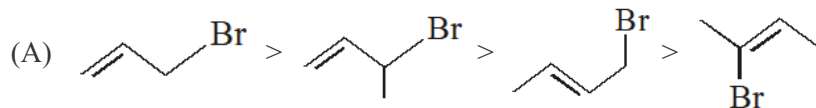
7. Find correct statements for Coordination compounds.
- (A) $[\text{PtCl}_2(\text{NH}_3)_2]$ & $[\text{Cu}(\text{NH}_3)_4]^{2+}$ are square planar complexes.
- (B) In $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$, geometrical isomerism does not exist while optical isomerism exists
- (C) C-O bond order : $[\text{Ni}(\text{CO})_4] > [\text{Co}(\text{CO})_4]^- > [\text{Fe}(\text{CO})_4]^{2-}$.
- (D) $[\text{Ni}(\text{Cl}_2)(\text{PPh}_3)_2]$ has two geometrical isomers.
8. In which of the following cases, the synergic bonding takes place at the π^* -orbital of the ligand.
- (A) $[\text{PtCl}_3(\text{C}_2\text{H}_4)]^-$ (B) $[\text{Ni}(\text{CO})_4]$
- (C) $[\text{Cr}(\text{C}_6\text{H}_6)_2]$ (D) $[\text{Fe}(\pi - \text{C}_5\text{H}_5)_2]$
9. Choose correct statements/Reactions.
- (A) Hydrogen peroxide (H_2O_2) is used for the purpose of bleaching clothes in process of laundry.
- (B) $\text{PbS(s)} + 4\text{H}_2\text{O}_2(\text{aq}) \rightarrow \text{Pb(s)} + \text{SO}_2 + 4\text{H}_2\text{O(l)} + \text{O}_2$
- (C) $2[\text{Fe}(\text{CN})_6]^{3-} + \text{H}_2\text{O}_2 + 2\text{OH}^- \rightarrow 2[\text{Fe}(\text{CN})_6]^{4-} + 2\text{H}_2\text{O} + \text{O}_2$
- (D) $2\text{MnO}_4^- + 3\text{H}_2\text{O}_2 \rightarrow 2\text{MnO}_2 + 3\text{O}_2 + 2\text{H}_2\text{O} + 2\text{OH}^-$

Space for Rough Work

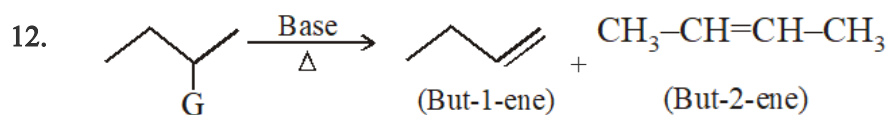
10. From which compound Grignard reagent can not be isolated when treated with Mg in ether?



11. Correct reactivity order towards S_N1 is/are :

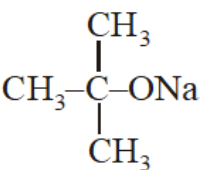


Space for Rough Work



But-1-ene is obtained as major product when :

(A) (G) is -F

(B) Base is 

(C) (G) is $\text{-}\overset{\oplus}{\text{N}}\text{Me}_3$

(D) (G) is -Cl & base is alc.KOH, Δ

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- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- Find the bond enthalpy (in kJ/mol) of one "three centre two electron bond" in B_2H_6 { $B-H-B \rightarrow 2B(g) + H(g)$ } from the given data
 $\Delta H_f^0 [BH_3(g)] = 100 \text{ kJ/mole}$ $\Delta H_f^0 [B_2H_6(g)] = 36 \text{ kJ/mole}$
 $\Delta H_{atm} [B(s)] = 565 \text{ kJ/mole}$ $\Delta H_{atm} [H_2(g)] = 436 \text{ kJ/mole}$
- An hypothetical element 'X' (Atomic weight = 20) undergo phase transformation : $X(s) \rightarrow X^*(s)$ at $25^\circ C$. Densities of X & X^* are respectively 2 gm/cc & 2.5 gm/cc. Find ΔG° (in J) if it attains equilibrium at 6 bar -
- | | |
|--|---|
| (i) $Ti^{+3} \Rightarrow$ Purple | (ii) $Fe^{+2} \Rightarrow$ Green |
| (iii) $Fe^{+3} \Rightarrow$ Yellow | (iv) $Ni^{+2} \Rightarrow$ Green |
| (v) $Cu^{+2} \Rightarrow$ Blue | (vi) $Sc^{3+} \Rightarrow$ Colourless |
| (vii) $Ti^{4+} \Rightarrow$ Colourless | (viii) $Zn^{2+} \Rightarrow$ Colourless |

How many colour of aqueous cation are correctly matched.
- Find the number of paramagnetic compound(s) with octahedral anion.
 $K_2[CoF_6]$; $O_2[AsF_6]$; $K_2[CoCl_4]$;
 $K_3[Co(CN)_6]$; $K_3[CoF_6]$; $[Ni(NH_3)_6]SO_4$;
 $[Pt(NH_3)_6][Fe(CN)_6]$; $Na_2[Fe^{II}(CN)_5NO]$

Space for Rough Work

(a) $\text{CH}_2 = \text{CH} - \text{Cl}$

(c) 

(d) 

(e) 

(g) 

(h) 

CC1(C)CCCC2(C)CCCC12Br >> [Alc.KOH][\Delta]

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options A, B, C and D. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Let $A = \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}$ and $(A + I)^{50} = 50A + \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $(2a + b + 4c + 3d)$ equals (where $a, b, c, d \in \mathbb{R}$)

- (A) 5 (B) 4 (C) 3 (D) 2

2. Consider a regular heptagon $A_1 A_2 A_3 A_4 A_5 A_6 A_7$ inscribed in a unit circle.

Let $p_1 = (A_1A_2)(A_1A_3)(A_1A_4)(A_1A_5)(A_1A_6)(A_1A_7)$

$p_2 = (A_2A_3)(A_2A_4)(A_2A_5)(A_2A_6)(A_2A_7)$

$p_3 = (A_3A_4)(A_3A_5)(A_3A_6)(A_3A_7)$

$p_4 = (A_4A_5)(A_4A_6)(A_4A_7)$

$p_5 = (A_5A_6)(A_5A_7)$ and $p_6 = A_6A_7$

where A_iA_j is length of line segment joining A_i and A_j then value of product $(p_1 p_2 p_3 p_4 p_5 p_6)$ is

- (A) 7^7 (B) $(\sqrt{7})^7$ (C) $(\sqrt{7})^6$ (D) $(\sqrt{7})^5$

Space for Rough Work

3. If $\left| \frac{z-2019}{z+2019} \right| = 1$ and $|z-3+4i| + |z-5-20i|$ is minimum, then the value of $\lim_{\theta \rightarrow 0} \frac{\sin(9z^2 + 30)\theta}{\tan^{-1}(z + \bar{z} + 2\theta)}$ is
- (A) $\frac{195}{2}$ (B) $\frac{195}{4}$ (C) $-\frac{195}{2}$ (D) $-\frac{195}{3}$
4. The number of positive integral solutions of the equation $x + y + z + w = 15$ where $x > 1$, $y > 2$, $z > 3$ and $w < 4$ is
- (A) 46 (B) 56 (C) 64 (D) 52

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Words are formed using all the letters of the word 'JEEADVANCED'.
- Let 'a' denotes the number of words in which all the vowels are together.
- 'b' denotes the number of words in which vowels as well as consonants are separated.
- 'c' denotes the number of words which begin and ends in vowels.
- Which of the following is(are) correct ?

(A) $c > a > b$

(B) $c > b > a$

(C) $a + b + c = 460.6!$

(D) $a + c = 91b$

Space for Rough Work

6. If A and B are two matrices such that $AB = A$ and $BA = B$, then
- (A) $(A^3 - B^3)^5 = (A^5 - B^5)^3$ (B) $(A^4 - B^4)^3 = A^{12} - B^{12}$
- (C) $(A - B)$ is idempotent (D) $(A^{10} - B^{10})$ is nilpotent
7. Parametric equation of a curve is given by
- $$x(t) = \frac{2 \cos(t)}{1 + a \sin(t)} ; y(t) = \frac{2 \sin(t)}{1 + a \sin(t)}$$
- Where a is real and positive. Then curve represents
- (A) an ellipse if $a^2 - 1 < 0$ (B) a parabola if $a^2 - 1 = 0$
- (C) a hyperbola if $a^2 - 1 > 0$ (D) a circle if $a^2 - 1 = 0$
8. Let z_1, z_2 and z_3 be three distinct complex numbers satisfying $|z_1| = |z_2| = |z_3| = 1$. Which of the following is / are true ?
- (A) If $|z_1 - z_2| = \sqrt{2} |z_1 - z_3| = \sqrt{2} |z_2 - z_3|$, then $\operatorname{Re}\left(\frac{z_3 - z_1}{z_3 - z_2}\right) = 0$
- (B) $|z_1 z_2 + z_2 z_3 + z_3 z_1| = |z_1 + z_2 + z_3|$
- (C) $\operatorname{Im}\left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 \cdot z_2 \cdot z_3}\right) = 0$
- (D) If $|z_1 - z_2| = \sqrt{2} |z_1 - z_3| = \sqrt{2} |z_2 - z_3|$, then $\operatorname{Re}\left(\frac{z_3 - z_1}{z_3 - z_2}\right) \neq 0$

Space for Rough Work

9. Let matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a, b, c, d \in \{1, 2, 3\}$ then
- (A) Total number of symmetric matrices are 30
- (B) Total number of singular matrices are 15
- (C) Total number of singular matrices are 21
- (D) Total number of symmetric matrices are 27
10. A parabola whose focus is $S(6, 8)$ is touching the co-ordinate axes, then which of the following statement(s) is/are correct ?
- (A) The equation of circle whose diameter is the portion of the tangent at the vertex of the parabola between the co-ordinate axes is $x^2 + y^2 - 6x - 8y = 0$.
- (B) If P and Q are the ends of a focal chord of the parabola, then $\frac{1}{SP} + \frac{1}{SQ} = \frac{5}{24}$.
- (C) Equation of the latus rectum of the parabola is $4x + 3y = 48$
- (D) If P and Q are the ends of a focal chord of the parabola, then $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{5}$

Space for Rough Work

11. For the conic given by $(x - 2y - 7)(x + 2y + 5) = 4$, then
- (A) equation of auxiliary circle is $x^2 + y^2 - 2x + 6y + 6 = 0$
- (B) equation of auxiliary circle is $x^2 + y^2 - 2x + 6y + 9 = 0$
- (C) locus of point of intersection of mutually perpendicular tangents is $x^2 + y^2 - 2x + 6y + 7 = 0$
- (D) locus of point of intersection of mutually perpendicular tangents is $x^2 + y^2 - 2x + 6y + 5 = 0$
12. Each of 2010 boxes in a line contains one red marble and for $1 \leq k \leq 2010$, the box in the k^{th} position also contains k white marbles. A child begins at the first box and successively draws a single marble at random from each box in order from 1st box to 2010th box. He stops when he first draws a red marble. Let $P(n)$ be the probability that he stops after drawing exactly 'n' marbles. The possible value(s) of n for which $P(n) < 1/2010$ is/are
- (A) 77 (B) 46 (C) 44 (D) 45

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the parabola $y^2 = 2x$. They intersect at P and Q in the first and fourth quadrant respectively. Tangent to the ellipse at P and Q intersect the x-axis at R and tangent to parabola at P and Q intersect the x-axis at S. If the area of quadrilateral PRQS is A then determine the value of $\left(\frac{2}{25\sqrt{3}}A\right)$
2. Let tangents be drawn to the curve, $y^2 - 4(x + y) = 3 \sin \theta + 4 \cos \theta - 15$, $\theta \in \mathbb{R}$ from the origin. It is given that the vertex of the curve is at maximum distance from the origin. If slopes of these tangents are m_1 and m_2 , then the value of $|m_1 m_2|$ equals
3. Three smallest squares are chosen randomly on a chess board. The probability that these squares have exactly two corners but no side common is $\frac{\lambda}{\mu}$ (where λ and μ are relatively prime numbers) then $\frac{\mu}{100}$ equals

Space for Rough Work

4. Tangents are drawn from any point on straight line $y = x + 8$ to the auxiliary circle of $x^2 + 16y^2 = 16$. If A and B are the points of contact of these tangents and P, Q are corresponding points of A and B on the ellipse respectively. If locus of mid point of PQ is $x^2 + \alpha y^2 + \beta x + \gamma y = 0$ then value of $\frac{(\alpha - \beta + \gamma)}{8}$ is
5. A biased coin with probability $p(0 < p < 1)$ of falling tails is tossed until a tail appear for the first time. If the probability that tail comes in odd number of trials is $\frac{2}{3}$, then p equals
6. Consider system of equations in x, y and z
 $12x + by + cz = 0$
 $ax + 24y + cz = 0$
 $ax + by + 36z = 0$
 (where a, b, c are real numbers, $a \neq 12$, $b \neq 24$, $c \neq 36$).
 If system of equation has solution and $z \neq 0$, then value of $5 \left| \frac{1}{a-12} + \frac{2}{b-24} + \frac{3}{c-36} \right|$ is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work